AUTOIMMUNITY AND IMMUNE TOLERANCE

Dr. Feyan M. Abdullah
Hawler Medical University
College of Medicine
Microbiology, Physiology and Genetic Dep.
Medical Microbiology Unit

SGL7

Overview

- Autoimmunity refers to immune reactions against self (auto) Ags.
- Immunologic tolerance is a state of unresponsiveness to an Ag that is induced by exposure of specific lymphocytes to that Ag.

Tolerance is a feature of adaptive immunity.

- Autoimmune disease is caused by **failure** of the tolerance processes.

Two broad mechanisms are work to prevent immune reactions against the body's own Ags.

1. Central tolerance
2. Peripheral tolerance

Central tolerance

- The principal mechanism of central tolerance is the Ag-induced deletion (death) of self-reactive T lymphocytes (in the thymus) and B lymphocytes (in bone marrow) during their maturation in central lymphoid organs.

Central Tolerance for T cells

Within the thymus, T cells mature and undergo differentiation expressing a **specific TCR**, **CD3, CD4+, or CD8+.**

- **Positive Selection**:- In the thymus cortex, those T cells that are able to make functional contacts with MHC class I and II will receive survival signals.
- **Negative Selection:-** The surviving cells move into the thymus medulla, where those that make strong contacts with self-Ags are killed by apoptosis and/or generation of CD4+ regulatory T cells.
- T cells that are weakly bind self-Ags survive, exit the thymus as mature naïve T cells, and migrate throughout the circulation and secondary lymphoid tissues surveying for foreign Ags.

- The **induction of tolerance to self-Ags** is regulated by the autoimmune regulator gene (AIRE).....negative selection.

- AIRE is a transcription factor expressed by the **medullary thymic epithelial cells (mTECs)** that drives the expression of many tissues-specific antigens (TSAs) typically expressed at the periphery.

- Mutations in AIRE **cause** a breakdown of central tolerance associated with decreased expression of self antigens in the thymus.

What about B cells (central tolerance).....??

- When immature B lymphocytes interact strongly with self Ags in the bone marrow, the B cells either change their receptor specificity (receptor editing) or are killed (deletion/apoptosis) or by anergy.

  - **Positive selection:** Only those cells that do generate a BCR are allowed to survive and mature.
  - **Negative selection:** Only B cells that do not strongly bind Ags are allowed to leave the bone marrow and, therefore, will be self-tolerant.

- Not all self Ags are present in the thymus or bone marrow, and hence lymphocytes bearing receptors for such **auto-Ags** escape into the periphery……. (escape central tolerance).

Peripheral tolerance of T cells

- Tolerance acquired outside the thymus.

- It is necessary because:-

i. Some self-Ags are not expressed in the thymus, therefore, some self-reactive T cells are not eliminated by negative selection.

ii. There are foreign Ags, such as those from commensal organisms on the barriers of our skin and gastrointestinal tract, that are harmless but immune response to these Ags would be pathogenic (i.e., a hypersensitivity reaction and **autoimmunity**).

Peripheral tolerance takes place in the peripheral tissues and lymph nodes, and consists in different immunologic mechanisms capable to control self-reactive lymphocytes escaped from central deletion.

Peripheral tolerance of mature T lymphocytes

This is mediated by :-

- ✓ **Deletion by apoptosis:-** T cells that recognize self Ags may receive signals that promote their death by apoptosis.
- ✓ **Anergy**:- When T cells recognize Ags without costimulation (B7 protein on a DC that interacts with the CD28 protein on the T cell) , the TCR complex may lose its ability to transmit activating signals.
- ✓ **Neglect or ignorance**:- when the Ag is effectively invisible to the immune system… how and why its not visible?

↓

Immunological ignorance can occur because the Ag is **sequestered in an avascular organ** such as the intact vitreous humour of the eye, also those found in testis.

- ✓ **Suppression:-** T cells that recognize self Ags may differentiate into **regulatory cells** in the thymus or peripheral tissues, in a process that is dependent on the transcription factor **FoxP3**. These cells bearing **CD4+T** mostly.

- These **regulatory cells** inhibit the activation and differentiation of naive T cells by contact-dependent mechanisms or by secreting cytokines that inhibit T cell responses such **as IL-10.**

- **Regulatory T cells** also express **CTLA-4**, which may bind to **B7** molecules on APCs and reduce their ability to activate T cells via **CD28.**

a checkpoint molecule:- cell surface receptors that can turn off activated T cells when they interact with their ligand (inhibitory mechanism).
Example:- CTLA-4 and PD-1

Defective regulatory T cell function ⟶ **Autoimmune diseases (AIDs)**

- **Regulatory T cells** consume **IL-2** (T cell growth factor).

CD4+CD25+ Foxp3+ Treg cells

Foxp3 mutations in humans are the cause of certain disorder called IPEX (immune dysregulation, polyendocrinopathy, enteropathy, X-linked syndrome) syndrome.

These individuals develope AIDs, such as type 1 diabetes mellitus, and inflammatory bowel disease.

What about peripheral B cell tolerance ???

Mature B lymphocytes that encounter self Ags in peripheral lymphoid tissues become incapable of responding to that Ag…**how?**

If B cells recognize a protein Ag but do not receive T cell help.

become anergic

surface IgM (BCR) is turned off and they become non sensitive to activation.

apoptosis

Engagement with inhibitory receptors thus preventing activation.

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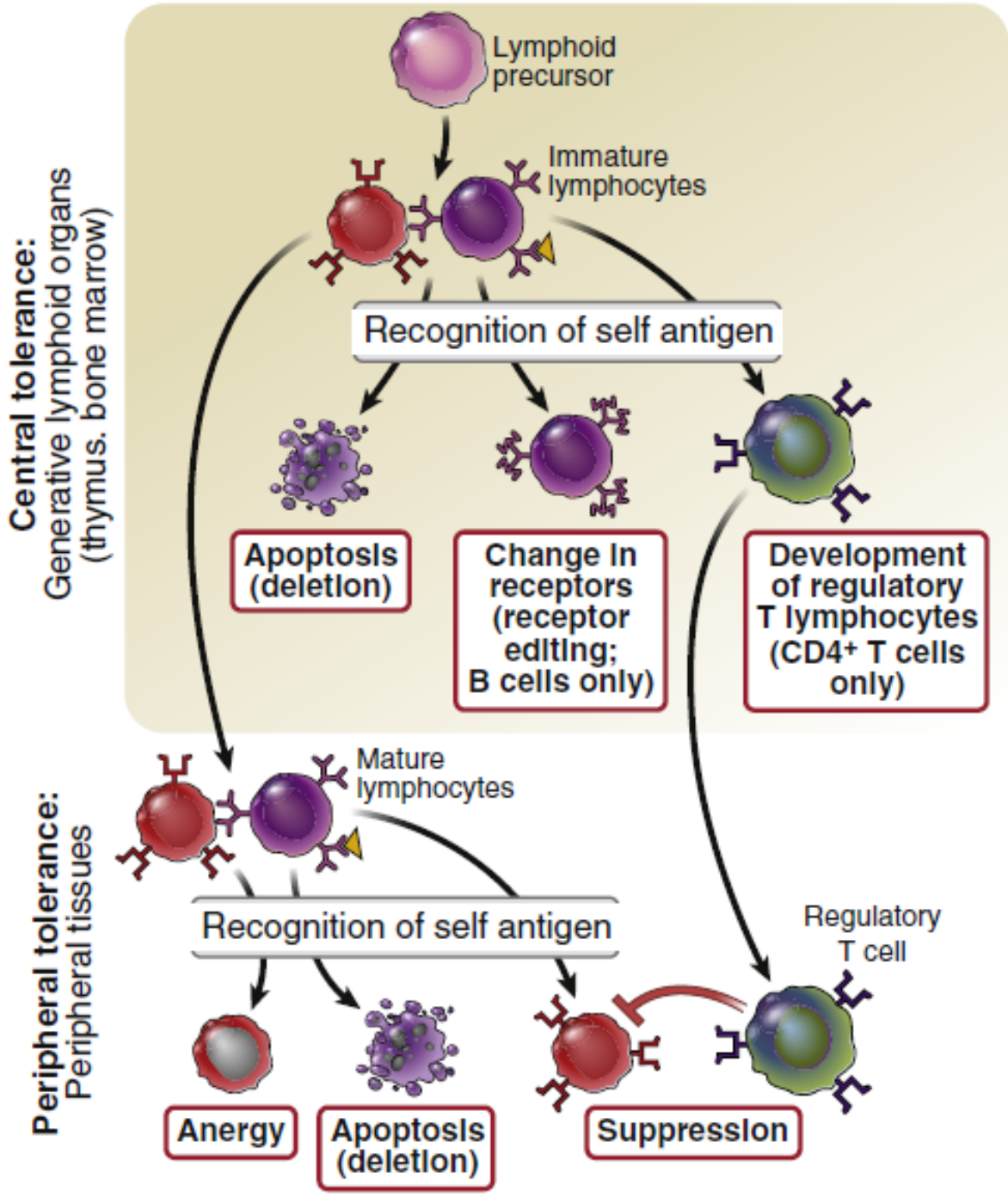


Fig. 9.1 Central and peripheral tolerance to self antigens.

What about tolerance to commensal microbes and fetal antigens

Where they found?

Mature lymphocytes in these tissues (eg. gut) are capable of recognizing the organisms but do not react against them…….**why and how?**

example

In the gut

Mechanisms ..examples:-

- Abundance of IL-10–producing regulatory T cells.
- Gut microbiome can influence the development of immune tolerance.

- Generation of peripheral FoxP3+ regulatory T cells specific for these paternal Ags.
- IL-10 is an anti-inflammatory cytokine promotes maternal-fetal tolerance by the suppression of pro-inflammatory responses and supporting Treg expansion
- Exclusion of inflammatory cells from the pregnant uterus.
- Poor Ag presentation in the placenta.
- Hormonal factor.

So that, autoimmune disease(AID) is caused by failure of the tolerance processes.

- AID can affect any organ in the body.
- In certain cases the damage to self cells or organs is caused by Abs; or by T cells, or by both.
- **Immune complexes**, **complement activation** other **inflammatory cells** and **cytokines** are involved in the process.
- However, the presence of an **auto-Abs** does not automatically mean that a person has AIDs (eg. ANA can be positive in healthy individuals, particularly at lower titers).
- AID tend to be chronic, progressive, and self perpetuating.....Epitope spreading
- AID can lead to morbidity and mortality from complications, and prolonged medical intervention.
- AID are often categorized to organ or **tissue specific** and **systemic** or multi-organ or tissue (non specific).

Auto immune diseases

Organ or tissue-specific

The **antigenic targets** may be molecules expressed on the surface of cells (eg. hormone receptors) or intracellular molecules, (eg. intracellular enzymes) .

example

Organ specific AID involve a single organ or tissue, or gland, such as:-

- **Thyroid gland** in Autoimmune thyroiditis (Hashimoto thyroiditis and Graves' disease.
- **The islets of Langerhans** in type I diabetes (T1DM).

Systemic (no organ or non tissue specific)

Affect **multiple organs** and are usually associated with autoimmune responses against self molecules that are widely distributed through the body, and particularly with intracellular molecules .

example

- **Systemic lupus erythematosus (SLE).**
- **Rheumatoid arthritis (RA).**

****Auto-Abs-mediated AIDs sometimes can be transmitted transplacentally……how and which type of Abs……..?

- **Example:-** Neonatal Graves' disease or congenital complete heart block and neonatal lupus.
- **IgG Abs** can cross the placenta.
- In most cases, neonatal AID is **transient**…..?

In view of the half-life of IgG, nearly all maternal IgG disappears from the circulation of the baby by six to twelve months postpartum.

- If maternal Abs cause irreversible organ damage in utero, does the damage disappear after birth? Like congenital heart block in neonatal lupus.
- The chance of developing AIDs after birth depend on what?

- Most **AID** involve a predominance of one or another form of hypersensitivity, but there is often considerable overlap between Ab and T-cell-mediated damage.

1. Type II hypersensitivity (caused by auto-Abs reactive with cell surface or matrix Ags).

2. Type III hypersensitivity (caused by immune complexes).

3. Type IV hypersensitivity (delayed-type hypersensitivity, mediated by T cells)

What triggers autoimmunity?

Interactions between genetic and environmental factors are critically important for autoimmunity.

- **Genetic Predisposition**

- Most AID are complex **multigenic** disorders.
- AID have a tendency to run in families, and there is a greater incidence of the same disease in monozygotic than in dizygotic twins.
- Several AID are linked to the HLA locus**.**

  RA ……HLA-DR4

  Ankylosing spondylitis ……HLA-B27.

- The occurrence of several AID in a given individual or family:-eg. hashimoto's thyroiditis is associated with a variety of **organ-specific** (e.g., T1DM, pernicious anemia), and **systemic** (e.g., SLE, RA) AID.

- **Non- MHC genes** also are strong candidates for AID susceptibility genes.

- **Polymorphisms** in the gene encoding **C2, C4** (Complement proteins) have been linked with **SLE**….. Defects in clearance of immune complexes.

- **Hormones**

- **Females** are more likely to be affected than **males**.
- Women are more prone to thyroiditis, SLE, multiple sclerosis (MS), and RA.....ankylosing spondylitis predominant in male.
- **Estrogen** and **prolactin** are immune stimulant while **androgen** are immune suppressor……eg. SLE exacerbates during pregnancy.

- **Release of sequestered Ags**……..trauma, surgery, infection can release these tissue-specific Ags → AID….**eg.** the testes, central nervous system, and the lens of the eye result in eliciting immune responses, producing aspermatogenesis, encephalitis, or endophthalmitis, respectively).

- **Environmental factors**

Infection, therapeutic drugs and other agents such as ultraviolet radiation are possible triggers in autoimmunity in susceptible hosts.

- **Infection**

Structural similarity (**Molecular mimicry**) between self-proteins and those from microorganisms (eg. Bacteria and viruses) may also trigger an autoimmune response.

✓ Rheumatic fever, which is thought to be a consequence of cross reactivity or molecular mimicry between Ags carried by group A *streptococci pyogen* and self-Ags of the heart**.**

- **Drugs**

Many drugs are associated with idiosyncratic side effects that may have an autoimmune pathogenesis. (eg. Procainamide-induced SLE).

- **Ultraviolet (UV) radiation** (usually in the form of sunlight) causes cell death and may lead to the exposure of nuclear Ags, which elicit pathologic immune responses in **SLE**.

- Lupus flares with exposure to sunlight.

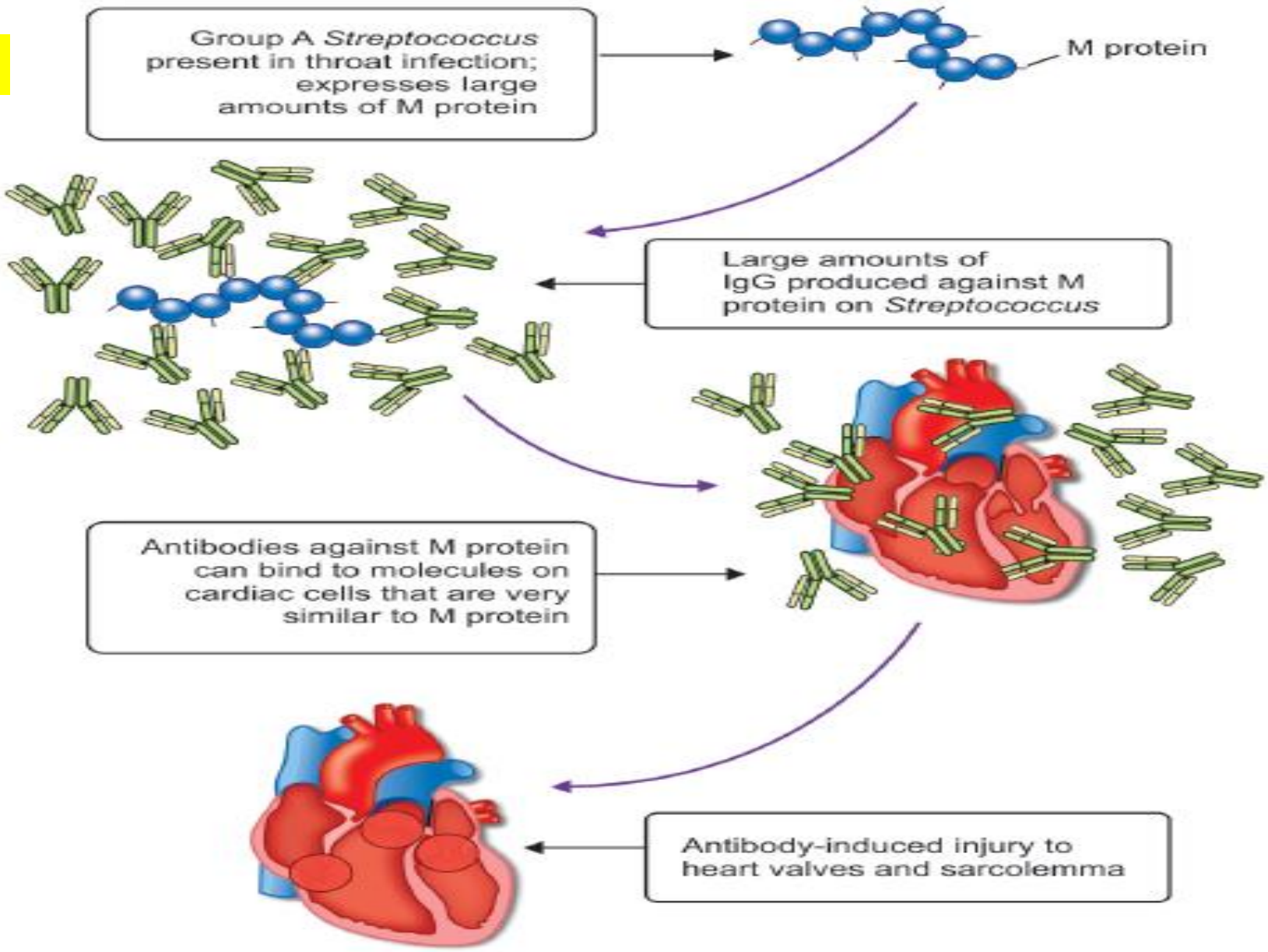
- Diet, obesity, smoking, and mucosal microflora, have all been associated with risk of AID, in susceptible genetic background.

- **Failure of Treg Cells :-**

✓ Tregs are CD4-positive cells that suppress the inflammatory effects of other T cells.

✓ **Mutations of the gene** encoding **AIRE** or the gene encoding **FOXP3** (required for Treg differentiation), cause autoimmunity through **deficiency** of Tregs.

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Rheumatic fever



FIGURE 188-4 Malar rash in an adolescent Hispanic girl with SLE. Note the relative sparing of the nasolabial folds. (*Reproduced with permission from Richard P. Usatine, MD.*)

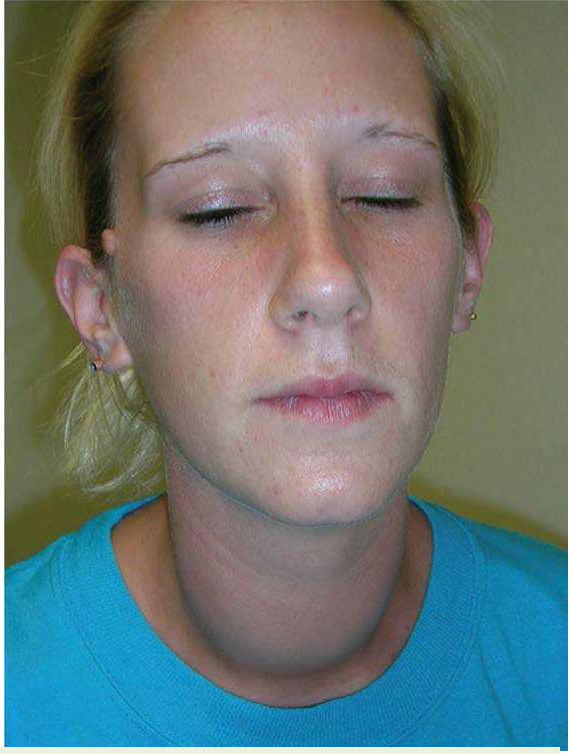


FIGURE 236-4 Enlarged thyroid, visible flushed skin and temporal wasting in this thyrotoxic woman with new-onset Graves disease. (Reproduced with permission from Richard P. Usatine, MD.)



FIGURE 99-5 Rheumatoid nodules in the hands. (Reproduced with permission from Richard P. Usatine, MD.)

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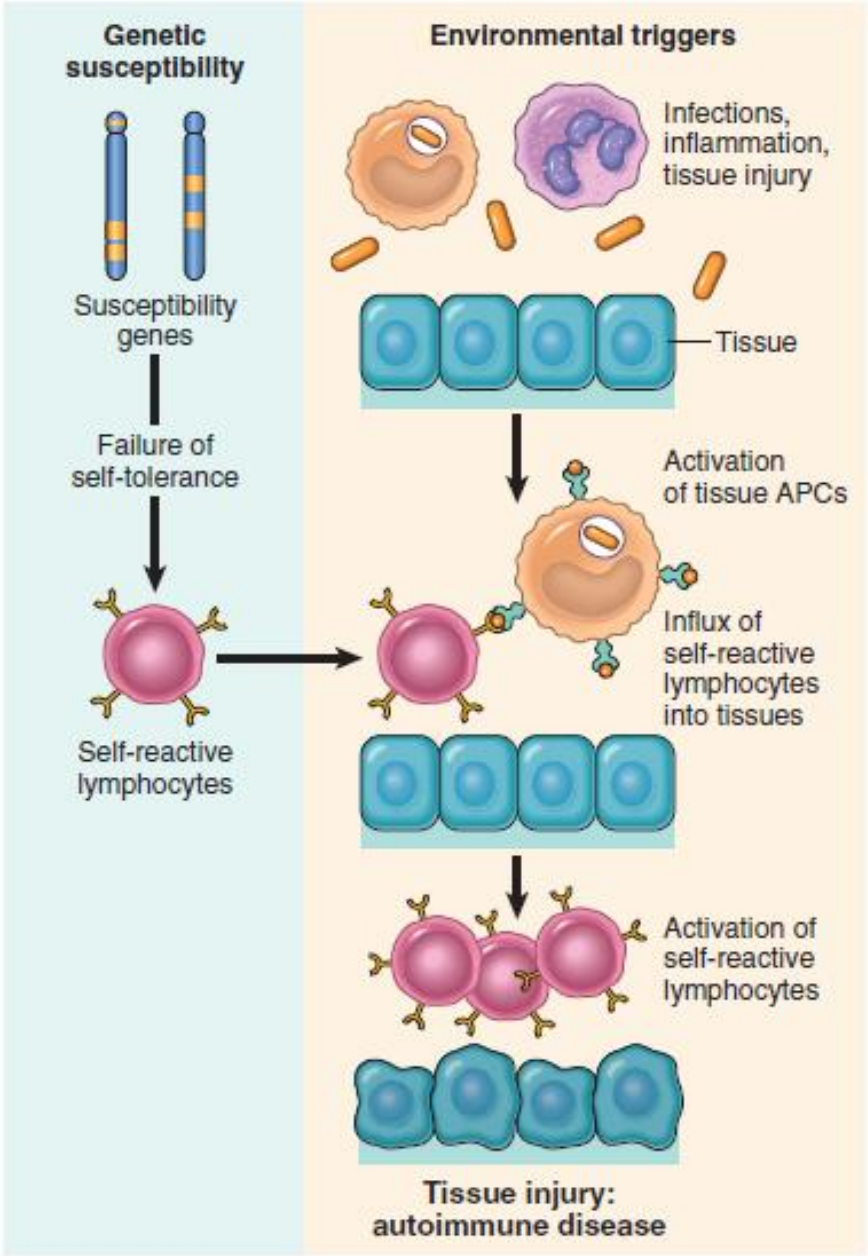


Fig. 5.20 Pathogenesis of autoimmunity. Autoimmunity results from multiple factors, including susceptibility genes that may interfere with self tolerance and environmental triggers (such as infections, tissue injury, and inflammation) that promote lymphocyte entry into tissues, activation of self-reactive lymphocytes, and tissue damage.

TABLE 16-1 Some autoimmune diseases in humans

Disease	Self antigen/Target gene	Immune effector
ORGAN-SPECIFIC AUTOIMMUNE DISEASES		
Addison's disease	Adrenal cells	Auto-antibodies
Autoimmune hemolytic anemia	RBC membrane proteins	Auto-antibodies **
Goodpasture's syndrome	Renal and lung basement membranes	Auto-antibodies
Graves' disease	Thyroid-stimulating hormone receptor	Auto-antibodies (stimulating) **
Hashimoto's thyroiditis	Thyroid proteins and cells	T_H1 cells, auto-antibodies**
Idiopathic thrombocytopenic purpura	Platelet membrane proteins	Auto-antibodies
Type 1 diabetes mellitus	Pancreatic beta cells	T_H1 cells, auto-antibodies **
Myasthenia gravis	Acetylcholine receptors	Auto-antibodies (blocking)**
Myocardial infarction	Heart	Auto-antibodies
Pernicious anemia	Gastric parietal cells; intrinsic factor	Auto-antibodies **
Poststreptococcal glomerulonephritis	Kidneys	Immune complexes
Spontaneous infertility	Sperm	Auto-antibodies

SYSTEMIC AUTOIMMUNE DISEASES		
Ankylosing spondylitis	Vertebrae	Immune complexes
Multiple sclerosis **	Brain or white matter	T_H1 cells and T_C cells, auto-antibodies
Rheumatoid arthritis **	Connective tissue, IgG	Auto-antibodies, immune complexes
Scleroderma	Nuclei, heart, lungs, gastrointestinal tract, kidneys	Auto-antibodies
Sjögren's syndrome	Salivary glands, liver, kidneys, thyroid	Auto-antibodies
*S*ystemic *l*upus *e*rythematosus (SLE) **	DNA, nuclear protein, RBC and platelet membranes	Auto-antibodies, immune complexes
*I*mmune dysregulation, *p*olyendocrinopathy, *e*nteropathy, *X*-linked (IPEX) syndrome	Multiorgan/loss of *FoxP3* gene	Missing regulatory T cells
*A*utoimmune *p*olyendocrine *s*yndrome type 1 (APS-1)	Multiorgan/loss of *AIRE* gene	Defective central tolerance

The diagnostic value of certain immune components in AIDs.

There is no single blood test that diagnoses all AIDs…

Testing should be guided by: **Clinical findings → suspected disease → appropriate test.**

- Antinuclear antibody (ANA)….. SLE (other AIDs may give positive also).
- Anti-dsDNA Abs……highly specific and sensitive for SLE.
- Anti-Sm Abs are highly specific for SLE but not highly sensitive.
- Complement components C1q, C2 and C4 ….SLE
- Rheumatoid factor (RF)….70% in RA
- Anticitrullinated protein Ab (ACCP)… high specificity for RA.
- Anti thyroid peroxidase(TPO) and anti thyroglobulin (TG) Abs ….. Hashimoto's Thyroiditis.
- Antibodies to thyroid stimulating hormone receptor (TSH- R)…Graves disease.
- Anti- acetylcholine receptor (AChRAb)…. Myasthenia gravis.
- Anti-tissue transglutaminase Abs (Anti-tTG IgA)….Important in Celiac disease.

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TABLE 66–3 Drugs Used in the Treatment of Autoimmune Diseases

Class of Drugs	Important Example of Drug	Mechanism of Action	Comment/Adverse Effect
Corticosteroids	Prednisone Methylprednisolone	1. Inhibits synthesis of inflammatory mediators such as IL-1 and TNF 2. Inhibits phospholipase, resulting in decreased arachidonic acid and prostaglandins 3. Induces apoptosis in lymphocytes	Used to control acute flares Long-term therapy limited by off-target toxicity on bones, endocrine systems
Nonsteroidal anti-inflammatory drugs (NSAIDs)	1. Aspirin 2. Ibuprofen, naproxen 3. Celecoxib	1. Inhibits cyclooxygenases (COX-1 and COX-2); this decreases prostaglandins 2. Same as aspirin 3. Inhibits only COX-2 so only inflammatory mediators are decreased	1. GI tract bleeding; Reye's syndrome 2. GI tract bleeding 3. Less GI tract bleeding than the NSAIDs described above
DNA synthesis inhibitors	Methotrexate	Inhibits DHFR, which decreases folate synthesis; this stops cell division of lymphocytes Suppresses T-cell and B-cell function	Bone marrow suppression; teratogenic
TNF inhibitors	1. Etanercept 2. Infliximab Adalimumab Certolizumab Golimumab	1. TNF receptor fused with an Fc region of an immunoglobulin; neutralizes TNF 2. Antibodies against TNF	Risk of activation of tuberculosis
Inhibitors of Th-1 and Th-17 pathway	1. Brodalumab 2. Ixekizumab Secukinumab 3. Ustekinumab	1. Antibody against the IL-17 receptor 2. Antibodies against IL-17 3. Antibody against IL-12 and IL-23	Different agents used for psoriasis, rheumatoid arthritis, ankylosing spondylitis, and Crohn's disease
Inhibitors of leukocyte migration	1. Natalizumab 2. Vedolizumab	1. Inhibits *all* α4 integrins 2. Inhibits α4β7 integrins (gut selective)	Progressive multifocal leukoencephalopathy Theoretical increased risk of GI infection
B-cell inhibitors	Rituximab Ocrelizumab	Both bind to CD20 on surface of B cells, resulting in apoptosis of B cells	PML and reactivation of HBV

CTLA = cytotoxic T-lymphocyte associated protein; DHFR = dihydrofolate reductase; GI = gastrointestinal; HBV = hepatitis B virus; IL = interleukin; PML= progressive multifocal leukoencephalopathy; TNF = tumor necrosis factor.

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